

CHEATSHEET · QUICK REFERENCE

Lab 29 — Troubleshooting Common Hardware Failures

Investigate event logs first, read SMART predictive attributes, map POST/LED/beep cues, decode crash-dump terms and separate host faults from misallocated-VM symptoms.

Event logs and sensors first

```
dmesg -T | tail -40
Kernel ring buffer

journalctl -k | grep -iE 'mce|edac|temperature|thrott|ecc'
Hardware error markers

sensors
CPU / board temperatures

stress-ng --cpu $(nproc) --timeout 10s --metrics-brief
Thermal load test
```

SMART predictive attributes

Attribute	Warns of
Reallocated_Sector_Ct	Bad sectors remapped
Current_Pending_Sector	Unstable sectors
Offline_Uncorrectable	Unrecoverable sectors
Media_Wearout_Indicator	SSD end of life

Crash dump terms + POST cues

Cue / term	Meaning
BSOD / PSOD / kernel panic	Windows / ESXi / Linux crash
Long repeating beep	Memory error
3 short beeps	Video / GPU
No POST + amber LED	PSU / power good
Burnt smell	VRM/cap failure - power off now

CMOS + VM misallocation

```
date ; chronyc tracking
Clock skew -> CMOS battery

top -bn1 | head
vCPU over-subscription

iostat -x 1 3
Storage over-subscription (high await)
```